

$$1. \ln \left( \frac{P_{\text{vap}}}{p^{\circ}} \right) = - \frac{\Delta H_{\text{vap}}}{RT} + C$$

molar vaporization enthalpy [ $\text{J} \cdot \text{mol}^{-1}$ ]  
 constant [dimensionless]  
 abs. temp [K]  
 formal pres. [1 atm]  
 vapor pres. [atm]

note: for dimensional consistency, logarithmic terms are often converted to dimensionless quantities by dividing by "formal" value, i.e. unit value.

$$\ln(P_{\text{vap}}) \rightarrow \ln \left( \frac{P_{\text{vap}}}{p^{\circ}} \right) \rightarrow \ln \left( \frac{[\text{atm}]}{[\text{atm}]} \right)$$

$$2. \underbrace{\ln \left( \frac{P_{\text{vap}}}{p^{\circ}} \right)}_y = \underbrace{\frac{-\Delta H_{\text{vap}}}{R}}_m \cdot \underbrace{\frac{1}{T}}_x + C$$



By measuring the  $P_{\text{vap}}$  at different known  $T$  values, we could determine  $\Delta H_{\text{vap}}$  (Gradient) and  $C$  (intercept)